MATH423: Introduction to Computational Topology
Midterm 1 Preparation Worksheet

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Coverage. Everything in the lecture notes up to and including Section 6.4: basic topology (Week 1), surfaces and triangulations (Week 2), abstract simplicial complexes, nerves, Čech and Vietoris–Rips complexes (Week 3), the interlude on applications (Section 4), the case studies of the computational lab (Section 5), and homology over $\mathbb{Z}/2$ up to the Euler–Poincaré formula (Sections 6.1–6.4).

How to use this worksheet. The midterm will be much shorter than this worksheet, and its questions will be of the same types. Try every problem before looking at the solutions at the end. Unless stated otherwise, homology has coefficients in $\mathbb{Z}/2$, and b_k denotes the k-th Betti number over $\mathbb{Z}/2$. For Rips complexes we use the convention of the notes: a set σ is a simplex of $VR_\epsilon(X)$ when all pairwise distances in σ are at most 2ϵ.

Part A: True or False

Decide whether each statement is true or false. Give a short proof if it is true and a counterexample if it is false.

A1. In every topological space, the intersection of any collection of open sets is open.

A2. Every metric space is Hausdorff.

A3. The open interval $(0, 1)$ is homeomorphic to $\mathbb{R}$.

A4. The intervals $[0, 1)$ and $(0, 1)$ are homeomorphic.

A5. A continuous bijection between topological spaces is always a homeomorphism.

A6. The continuous image of a connected space is connected.

A7. The figure eight (two circles joined at a point) is a 1-manifold.

A8. A closed connected surface with Euler characteristic 1 can be orientable.

A9. The square with boundary word $abab$ is the Klein bottle.

A10. Every triangulation of a closed surface has an even number of triangles.

A11. The star of a vertex in a simplicial complex is always a subcomplex.

A12. For every finite set $X \subset \mathbb{R}^n$ and every $\epsilon > 0$, $\check{C}_\epsilon(X) \subseteq VR_\epsilon(X)$.

A13. If $\check{C}_\epsilon(X)$ has a 1-dimensional hole, then so does $VR_\epsilon(X)$.

A14. Over $\mathbb{Z}/2$, every k-boundary is a k-cycle.

A15. Over $\mathbb{Z}/2$, every k-cycle is a k-boundary.

A16. If K is connected and has n_0 vertices, then $\operatorname{rank} \partial_1 = n_0 - 1$.

A17. Two finite simplicial complexes with the same Euler characteristic have the same Betti numbers.

A18. If K is a non-empty complex of dimension at most 2 with $b_1 = 0$, then $\chi(K) \geq 1$.

A19. If all pairwise distances in a finite set X are at most 2ϵ, then $VR_\epsilon(X)$ has the homology of a point.

A20. The torus and the Klein bottle have the same Betti numbers over $\mathbb{Z}/2$.

Part B: Computations

Topology (Week 1)

B1. Compute d_1, d_2 and d_∞ between $x = (-1, 3)$ and $y = (2, -1)$ in $\mathbb{R}^2$, and check that $d_\infty \leq d_2 \leq d_1 \leq \sqrt{2}\, d_2$.

B2. Which of the following are topologies on $X = \{a, b, c\}$?

$$\mathcal{T}_1 = \{\emptyset, \{a\}, \{b, c\}, X\}, \quad \mathcal{T}_2 = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}, \quad \mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}.$$

For each topology, list the closed sets.

B3. In $X = [0, 2]$ with the subspace topology from $\mathbb{R}$, decide whether each set is open in X, closed in X, both, or neither: $[0, 1)$, $(1, 2]$, $\{1\}$, $(1/2, 3/2)$, X.

B4. Give an explicit homeomorphism from $(0, 1)$ to $(1, \infty)$, and from $(-1, 1)$ to $\mathbb{R}$.

B5. Find an open cover of $[0, 1)$ with no finite subcover. Why does this not contradict the Heine–Borel theorem?

Surfaces (Week 2)

B6. For each hexagon word, find the vertex classes, compute χ, decide orientability, and identify the surface: (a) $abcabc$; (b) $abca^{-1}b^{-1}c$; (c) $abcb^{-1}a^{-1}c^{-1}$.

B7. Identify the surface $T^2 \# \mathbb{R}P^2 \# \mathbb{R}P^2$ in the classification theorem.

B8. A closed connected surface is triangulated with 8 vertices and 16 triangles. How many edges does the triangulation have? Which surfaces are possible?

B9. Show that every triangulation of a closed surface with Euler characteristic χ and V vertices satisfies $V^2 - 7V + 6\chi \geq 0$. Deduce lower bounds for the number of vertices of a triangulation of S^2, of T^2 and of the orientable surface of genus 2.

B10. Compute the Euler characteristic of a sphere with 3 holes (that is, S^2 with the interiors of 3 disjoint disks removed), and of a torus with 2 holes.

B11. Which closed surfaces contain a Möbius band? Answer for S^2, T^2, $\mathbb{R}P^2$, the Klein bottle and $T^2 \# T^2$.

B12. Decide which of the following are surfaces (possibly with boundary): a closed disk; two closed disks glued at one boundary point; the union of the xy-plane and the z-axis in $\mathbb{R}^3$; an open cylinder $S^1 \times (0, 1)$.

B13. Let K be the triangulation of a closed surface. Explain why the link of every vertex is a cycle with at least 3 vertices, and why every vertex has degree at least 3.

Simplicial complexes and complexes from data (Week 3)

B14. Find the barycentric coordinates of $(1, 1)$, $(\frac{3}{2}, \frac{3}{2})$ and $(2, 2)$ with respect to the triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 3)$. Which points lie inside, on the boundary, or outside the triangle?

B15. Is the family $\{\{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$ an abstract simplicial complex? If not, find the smallest complex containing it.

B16. Let K be the complex with maximal simplices $\{1, 2, 3\}$, $\{1, 3, 4\}$, $\{4, 5\}$, $\{5, 6\}$, $\{4, 6\}$. Find (a) the f-vector and $\chi(K)$; (b) the star, closed star and link of the vertex 1; (c) the link of the edge $\{1, 3\}$; (d) the link of the vertex 4.

B17. List all simplices of the full 3-simplex Δ^3 and verify that $\chi(\Delta^3) = 1$. What is χ of the 1-skeleton of Δ^3?

B18. Cover the circle S^1 by four closed arcs, each overlapping only its two neighbors. Describe the nerve and compare its homology with that of S^1.

B19. Let $X = \{(0, 0), (4, 0), (4, 3), (0, 3)\}$. Describe $VR_\epsilon(X)$ and $\check{C}_\epsilon(X)$ for all $\epsilon > 0$. For which ϵ is the complex homotopy equivalent to a circle?

B20. Let X be the vertices of an equilateral triangle with side 2. Describe $\check{C}_{1.1}(X)$ and $VR_{1.1}(X)$, and verify $\check{C}_{1.1}(X) \subseteq VR_{1.1}(X) \subseteq \check{C}_{1.1\sqrt{2}}(X)$.

B21. Let X be the six vertices of a regular hexagon with side 1. Describe $VR_\epsilon(X)$ for $\epsilon = 0.4$, 0.6, 0.9 and 1, and give its f-vector in each case.

B22. How many simplices does $VR_\epsilon(X)$ have if X has N points, all at pairwise distance at most 2ϵ? How many triangles if $N = 100$?

Homology over $\mathbb{Z}/2$ (Sections 6.1–6.4)

B23. In the boundary of the tetrahedron on $\{1,2,3,4\}$, compute $\partial_2(\{1,2,3\}+\{1,2,4\}+\{1,3,4\})$ and then ∂_1 of the result.

B24. Write the boundary matrix ∂_1 of the path $1-2-3-4$ and compute b_0 and b_1.

B25. Let K be the square $\{1,2,3,4\}$ with the diagonal $\{1,3\}$ and the two triangles $\{1,2,3\}$, $\{1,3,4\}$. Decide whether each chain is a cycle, and whether it is a boundary: (a) $\{1,2\}+\{2,3\}+\{3,4\}+\{1,4\}$; (b) $\{1,2\}+\{2,3\}$; (c) $\{1\}+\{3\}$; (d) $\{1\}+\{2\}+\{3\}$.

B26. Compute the Betti numbers of: (a) the 1-skeleton of the tetrahedron; (b) the complete graph K_5; (c) the 1-skeleton of the octahedron.

B27. Compute the Betti numbers of two hollow triangles sharing one vertex, and give a basis of H_1.

B28. Compute the Betti numbers of the complex with maximal simplices $\{1,2,3\}$, $\{3,4\}$, $\{4,5\}$, and of the complex in Problem B16.

B29. Let K have the maximal simplices $\{1,2,3\}$, $\{4,5\}$ and $\{6\}$. Compute $\operatorname{rank}\partial_1$, $\operatorname{rank}\partial_2$ and all Betti numbers. Give a basis of H_0.

B30. Compute the Betti numbers of the triangulated annulus with triangles $\{1,2,4\}$, $\{2,4,5\}$, $\{2,3,5\}$, $\{3,5,6\}$, $\{1,3,6\}$, $\{1,4,6\}$. Show that the inner loop $\{1,2\}+\{2,3\}+\{1,3\}$ and the outer loop $\{4,5\}+\{5,6\}+\{4,6\}$ are homologous.

B31. Compute the Betti numbers of (a) the boundary of the octahedron; (b) the boundary of the octahedron with one triangle removed; (c) the cone on a hexagon, with triangles $\{0,k,k+1\}$ for $k=1,\dots,6$ (indices mod 6).

B32. Compute the Betti numbers of two boundaries of tetrahedra glued (a) at a vertex; (b) along an edge.

B33. Use Proposition 6.14 and the Euler–Poincaré formula to find the Betti numbers of: the 9-vertex torus; the 9-vertex torus with one triangle removed; the orientable surface of genus 3; the connected sum of 5 projective planes.

B34. For the hexagon of Problem B21, compute the Betti numbers of $VR_\epsilon(X)$ for $\epsilon = 0.6$, 0.9 and 1.

B35. A connected 2-dimensional complex has $n=(7,15,9)$ and $b_2=0$. Find b_1. A classmate claims the answer $b_1=2$; explain why this is impossible without any further computation.

B36. Let K be the 2-skeleton of the 4-simplex (all vertices, edges and triangles on 5 vertices). Show that $b_1=0$ and deduce b_2 from the Euler–Poincaré formula.

Part C: Proofs

C1. Prove that open balls in a metric space are open, and that the intersection of two open sets in a metric space is open.

C2. Prove that the continuous image of a compact space is compact.

C3. Prove that S^1 is not homeomorphic to $\mathbb{R}$.

C4. Let K be a triangulation of a closed surface with V vertices, E edges and F triangles. Prove that $3F = 2E$, $E = 3(V - \chi)$ and $F = 2(V - \chi)$. Deduce that every triangulation of the torus has at least 7 vertices.

C5. Prove that $\chi(S_1 \# S_2) = \chi(S_1) + \chi(S_2) - 2$.

C6. Prove that the nerve of a finite collection of sets is an abstract simplicial complex.

C7. Prove that $\check{C}_\epsilon(X) \subseteq VR_\epsilon(X)$ for every finite subset X of a metric space.

C8. Prove that the star of a simplex σ is a subcomplex only if σ is a vertex contained in no other simplex.

C9. Prove that $\partial_{k-1} \circ \partial_k = 0$ over $\mathbb{Z}/2$, and deduce that $B_k \subseteq Z_k$.

C10. Prove that b_0 equals the number of connected components of the 1-skeleton.

C11. Prove the Euler–Poincaré formula $\sum_k (-1)^k n_k = \sum_k (-1)^k b_k$.

C12. Prove that a triangulated closed connected surface has $b_2 = 1$ over $\mathbb{Z}/2$. Deduce $b_1 = 2 - \chi$.

C13. Show that $b_k(K \sqcup L) = b_k(K) + b_k(L)$ for the disjoint union of two complexes.

Part D: Applications and Concepts (Sections 4 and 5)

D1. Give two reasons why computational topology represents spaces by simplicial complexes rather than by graphs, and two reasons why it prefers abstract complexes to geometric ones.

D2. A program builds VR_ϵ of the four corners of the unit square with $\epsilon = 0.75$, but only up to dimension 2. What Euler characteristic does it report, and what is the correct one? Explain the general lesson.

D3. Check the link condition $\mathrm{lk}(a) \cap \mathrm{lk}(b) = \mathrm{lk}(\{a, b\})$ for an edge of the octahedron and for an edge of the tetrahedron. What does the result mean for edge contraction?

D4. Prove that a triangle in the plane with all sides at most r and all angles at most 90° is covered by the three disks of radius $r/\sqrt{3}$ centered at its vertices. Why is this relevant for sensor networks?

D5. The Stanford bunny mesh has 34,834 used vertices, 104,288 edges, 69,451 triangles, 223 boundary edges in 5 loops, and is connected and orientable. Verify $3F = 2E - B$, compute χ, and determine the genus. What would the Euler characteristic be after capping each hole with a cone?

D6. In the toy protein of Section 5 (11 atoms of radius 1.7 Å on a circle of radius 6 Å, one atom missing), explain why the pocket becomes an enclosed cavity at inflated radius $R = 3$ Å and is filled at $R = 6$ Å. For which probe radii is there a cavity?

D7. Explain why a single failed interior sensor in a square grid with communication radius $\sqrt{2}$ (spacing 1) creates a 1-dimensional hole in the Rips complex, while a failed corner sensor does not. Is the corner still covered if $r_c = \sqrt{2}/\sqrt{3}$?

Solutions

Part A

A1. False. In $\mathbb{R}$, $\bigcap_{n\geq 1}(-\frac{1}{n},\frac{1}{n}) = \{0\}$, which is not open. Only finite intersections need to be open.

A2. True. If $x \neq y$, put $r = d(x,y)/2$. The balls $B_r(x)$ and $B_r(y)$ are disjoint: a common point z would give $d(x,y) \leq d(x,z) + d(z,y) < 2r = d(x,y)$.

A3. True. For example $t \mapsto \tan(\pi(t - \frac{1}{2}))$, or the composition of $t \mapsto 2t - 1$ with $s \mapsto \frac{s}{1-|s|}$.

A4. False. Removing the point 0 from $[0,1)$ leaves a connected space, but removing any point p from $(0,1)$ leaves the disconnected space $(0,p) \cup (p,1)$. A homeomorphism f would restrict to a homeomorphism $[0,1) \setminus \{0\} \to (0,1) \setminus \{f(0)\}$.

A5. False. $f : [0, 2\pi) \to S^1$, $f(t) = (\cos t, \sin t)$ is a continuous bijection whose inverse is not continuous at $(1,0)$.

A6. True. If $f : X \to Y$ is continuous and $f(X) = U \cup V$ with U, V disjoint, non-empty and open in $f(X)$, then $X = f^{-1}(U) \cup f^{-1}(V)$ is a decomposition of X into disjoint non-empty open sets, contradicting the connectedness of X.

A7. False. A small connected neighborhood of the crossing point, with that point removed, has four components; an interval minus a point has two.

A8. False. An orientable closed surface has $\chi = 2 - 2g$, which is even. A closed surface with $\chi = 1$ is $\mathbb{R}P^2$.

A9. False. It is $\mathbb{R}P^2$: the vertex classes are $\{P_0, P_2\}$ and $\{P_1, P_3\}$, so $\chi = 2 - 2 + 1 = 1$. The Klein bottle is given by $aabb$ or $aba^{-1}b$.

A10. True. $3F = 2E$ is even and 3 is odd, so F is even.

A11. False. In the boundary of the tetrahedron, $\mathrm{st}(1)$ contains $\{1,2,3\}$ but not its face $\{2,3\}$.

A12. True. See C7.

A13. False. For an equilateral triangle with side 2 and $\epsilon = 1.1$, $\check{C}_\epsilon$ is a hollow triangle but VR_ϵ is a solid triangle (Problem B20).

A14. True. If $c = \partial_{k+1} d$, then $\partial_k c = \partial_k \partial_{k+1} d = 0$.

A15. False. In the hollow triangle, $\{1,2\} + \{2,3\} + \{1,3\}$ is a cycle, but $B_1 = 0$ since there are no triangles.

A16. True. By Theorem 6.12, $b_0 = n_0 - \operatorname{rank} \partial_1$ (as $\partial_0 = 0$) equals the number of components, which is 1.

A17. False. The boundary of the tetrahedron and two isolated points both have $\chi = 2$, but their Betti numbers are $(1,0,1)$ and (2).

A18. True. $\chi = b_0 - b_1 + b_2 = b_0 + b_2 \geq b_0 \geq 1$.

A19. True. $VR_\epsilon(X)$ is then the full simplex on X, which is a cone; its Betti numbers are $(1, 0, 0, \dots)$.

A20. True. Both are closed connected surfaces with $\chi = 0$, so by Proposition 6.14 both have $b = (1, 2, 1)$ over $\mathbb{Z}/2$. (Integer coefficients distinguish them, but these come after Section 6.4.)

Part B

B1. The coordinate differences are 3 and 4, so $d_1 = 7$, $d_2 = 5$, $d_\infty = 4$, and $4 \leq 5 \leq 7 \leq 5\sqrt{2} \approx 7.07$.

B2. $\mathcal{T}_1$ is a topology, with closed sets $X, \{b, c\}, \{a\}, \emptyset$ (every open set is also closed). $\mathcal{T}_2$ is a topology, with closed sets $X, \{b, c\}, \{a, c\}, \{c\}, \emptyset$. $\mathcal{T}_3$ is not a topology: $\{a, b\} \cap \{b, c\} = \{b\} \notin \mathcal{T}_3$.

B3. $[0, 1) = (-1, 1) \cap X$ is open, not closed. $(1, 2] = (1, 3) \cap X$ is open, not closed. $\{1\}$ is closed, not open. $(\frac{1}{2}, \frac{3}{2})$ is open, not closed. X is both.

B4. $t \mapsto 1/t$ maps $(0, 1)$ onto $(1, \infty)$, with inverse $s \mapsto 1/s$. For $(-1, 1) \to \mathbb{R}$ use $t \mapsto \frac{t}{1-|t|}$, with inverse $s \mapsto \frac{s}{1+|s|}$.

B5. $U_n = [0, 1 - \frac{1}{n})$, $n \geq 2$, are open in $[0, 1)$ and cover it, but finitely many of them have union $[0, 1 - \frac{1}{N})$, which misses $1 - \frac{1}{2N}$. No contradiction: $[0, 1)$ is bounded but not closed in $\mathbb{R}$.

B6. Label the corners $P_0, \dots, P_5$. (a) *abcabc*: the identifications are $P_0 \sim P_3$, $P_1 \sim P_4$, $P_2 \sim P_5$, so $V = 3$, $E = 3$, $\chi = 1$; every letter appears twice with the same sign, so it is non-orientable: $\mathbb{R}P^2$. (Each boundary point is glued to its antipode.) (b) $abca^{-1}b^{-1}c$: all six corners are identified, $V = 1$, $E = 3$, $\chi = -1$; c appears twice with the same sign, so it is non-orientable: the connected sum of 3 projective planes. (c) $abcb^{-1}a^{-1}c^{-1}$: two vertex classes $\{P_0, P_2, P_3, P_5\}$ and $\{P_1, P_4\}$, so $\chi = 2 - 3 + 1 = 0$; every letter appears with both signs, so it is orientable: the torus.

B7. It is non-orientable with $\chi = 0 + 1 + 1 - 2 - 2 = -2 = 2 - k$, so $k = 4$: the connected sum of 4 projective planes (consistent with Dyck's theorem, $2g + k = 2 + 2$).

B8. $E = 3F/2 = 24$, so $\chi = 8 - 24 + 16 = 0$: the torus or the Klein bottle.

B9. $E = 3(V - \chi)$ (Problem C4) and $E \leq \binom{V}{2}$ give $3V - 3\chi \leq \frac{V(V-1)}{2}$, that is, $V^2 - 7V + 6\chi \geq 0$, so $V \geq \frac{7+\sqrt{49-24\chi}}{2}$, since the other root is at most 3 while $V \geq 4$ (the link of a vertex is a cycle with at least 3 vertices). For S^2: $V \geq 4$. For T^2: $V \geq 7$. For genus 2, $\chi = -2$: $V \geq \frac{7+\sqrt{97}}{2} \approx 8.42$, so $V \geq 9$. (The first two bounds are attained; the third is not always attained, but this is beyond the course.)

B10. Removing an open disk lowers χ by 1. Sphere with 3 holes: $2 - 3 = -1$. Torus with 2 holes: $0 - 2 = -2$.

B11. Exactly the non-orientable ones: $\mathbb{R}P^2$ and the Klein bottle. S^2, T^2 and $T^2 \# T^2$ are orientable.

B12. The closed disk is a surface with boundary. Two disks glued at a boundary point are not: the common point has no neighborhood homeomorphic to an open disk or a half-disk (removing it disconnects small neighborhoods into two pieces). The plane together with the z-axis is not: points of the z-axis other than the origin have small neighborhoods homeomorphic to an open interval, which is not homeomorphic to an open disk. The open cylinder is a surface without boundary.

B13. Every edge lies in exactly two triangles, so in the link of a vertex v every vertex w has degree 2 (the two triangles containing the edge vw). Worksheet 1, Problem 6 shows that the link is connected, so it is a single cycle. A cycle in a simplicial complex has at least 3 vertices, since two vertices span at most one edge. The degree of v is the number of vertices of its link, hence at least 3.

B14. From $x = (3t_1, 3t_2)$: $(1,1)$ has $(t_0, t_1, t_2) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, the barycenter, inside. $(\frac{3}{2}, \frac{3}{2})$ has $(0, \frac{1}{2}, \frac{1}{2})$, on the edge spanned by $(3,0)$ and $(0,3)$. $(2,2)$ has $(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$, outside.

B15. No: $\{1,3\}$ is a face of $\{1,2,3\}$ but is missing. Adding $\{1,3\}$ gives the smallest complex containing the family: a triangle with all its faces, together with the isolated vertex 4.

B16. (a) Vertices $1,\dots,6$; edges $12, 13, 23, 14, 34, 45, 56, 46$; triangles $123, 134$. So $f = (6, 8, 2)$ and $\chi = 0$. (b) $\mathrm{st}(1) = \{1, 12, 13, 14, 123, 134\}$. The closed star adds $2, 3, 4, 23, 34$, giving 11 simplices. $\mathrm{lk}(1) = \{2, 3, 4, 23, 34\}$, the path $2 - 3 - 4$. (c) $\mathrm{lk}(\{1,3\}) = \{2, 4\}$, two points (the edge $\{1,3\}$ lies in two triangles). (d) The simplices containing 4 are $4, 14, 34, 45, 46, 134$. So $\mathrm{lk}(4) = \{1, 3, 5, 6, 13\}$: the edge 13 together with the two isolated points 5 and 6.

B17. Δ^3 has 4 vertices, 6 edges, 4 triangles and 1 tetrahedron: $\chi = 4 - 6 + 4 - 1 = 1$. Its 1-skeleton has $\chi = 4 - 6 = -2$.

B18. The nerve has 4 vertices and 4 edges forming a cycle (a hollow square), and no triangles, since no three arcs meet. Its Betti numbers are $(1,1)$, those of S^1, as predicted by the Nerve Theorem: all non-empty intersections are arcs, hence contractible.

B19. The distances are 3 (short sides), 4 (long sides) and 5 (diagonals). Any three points form a right triangle with hypotenuse 5, whose smallest enclosing ball has radius $\frac{5}{2}$, and the ball of radius $\frac{5}{2}$ around $(2, \frac{3}{2})$ contains all four points. So both complexes change at the same values: four points for $\epsilon < \frac{3}{2}$; two disjoint edges for $\frac{3}{2} \le \epsilon < 2$; the hollow rectangle for $2 \le \epsilon < \frac{5}{2}$; the full 3-simplex for $\epsilon \ge \frac{5}{2}$. Homotopy equivalent to a circle exactly for $2 \le \epsilon < \frac{5}{2}$.

B20. All distances are $2 \le 2.2$, so $VR_{1.1}(X)$ is the solid triangle. The circumradius is $2/\sqrt{3} \approx 1.155 > 1.1$, and the triangle is acute, so the three closed disks of radius 1.1 have no common point while each pair meets (as $2 \le 2.2$): $\check{C}_{1.1}(X)$ is the hollow triangle. Since $1.1\sqrt{2} \approx 1.556 > 1.155$, $\check{C}_{1.1\sqrt{2}}(X)$ is the solid triangle. So the inclusions read: hollow $\subseteq$ solid $\subseteq$ solid.

B21. The distances are 1, $\sqrt{3} \approx 1.73$ and 2, so simplices appear at $2\epsilon = 1$, $\sqrt{3}$, 2. For $\epsilon = 0.4$: six points, $f = (6)$. For $\epsilon = 0.6$: the hexagon, $f = (6, 6)$. For $\epsilon = 0.9$: all pairs except the three opposite pairs; the simplices are the sets containing no opposite pair, which form the boundary of the octahedron, $f = (6, 12, 8)$. For $\epsilon = 1$: the full 5-simplex, $f = (6, 15, 20, 15, 6, 1)$.

B22. All non-empty subsets: $2^N - 1$ simplices. For $N = 100$ there are $\binom{100}{3} = 161{,}700$ triangles.

B23. The edges $12, 13, 14$ each appear twice and cancel: $\partial_2(\ldots) = \{2,3\} + \{2,4\} + \{3,4\}$. Then ∂_1 of this is $(2+3) + (2+4) + (3+4) = 0$.

B24. With edges $12, 23, 34$: $\partial_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, of rank 3. So $b_0 = 4 - 3 = 1$ and $b_1 = 3 - 3 = 0$.

B25. (a) A cycle, and the boundary of $\{1,2,3\} + \{1,3,4\}$. (b) Not a cycle: its boundary is $\{1\} + \{3\}$. (c) Every 0-chain is a cycle, and $\{1\} + \{3\} = \partial_1\{1,3\}$ is a boundary. (d) A cycle, but not a boundary: every 0-boundary has an even number of vertices.

B26. For a connected graph, $b_0 = 1$ and $b_1 = E - V + 1$. (a) $6-4+1 = 3$. (b) $10-5+1 = 6$. (c) $12 - 6 + 1 = 7$.

B27. $n = (5, 6)$, connected, so $\operatorname{rank}\partial_1 = 4$, $b_0 = 1$, $b_1 = 6 - 4 = 2$. A basis of H_1: the two triangle loops. Since there are no triangles, $B_1 = 0$, so distinct cycles are distinct classes.

B28. First complex: $n = (5, 5, 1)$, connected, $\operatorname{rank}\partial_1 = 4$, $\operatorname{rank}\partial_2 = 1$: $b = (1, 0, 0)$. (It deformation retracts to a point.) Complex of B16: $n = (6, 8, 2)$, $\operatorname{rank}\partial_1 = 5$, $\operatorname{rank}\partial_2 = 2$ (the two triangles have independent boundaries): $b = (1, 8-5-2, 0) = (1, 1, 0)$. The hole is the hollow triangle $4, 5, 6$. Check: $\chi = 0 = 1 - 1 + 0$.

B29. $n = (6, 4, 1)$. Three components, so $\operatorname{rank}\partial_1 = 6 - 3 = 3$; $\operatorname{rank}\partial_2 = 1$. Hence $b = (3, 0, 0)$, and a basis of H_0 is $\{1\}, \{4\}, \{6\}$.

B30. $n = (6, 12, 6)$, $\operatorname{rank}\partial_1 = 5$. A 2-cycle can contain no triangle with a boundary edge, and every triangle has one, so $\ker\partial_2 = 0$ and $\operatorname{rank}\partial_2 = 6$. Hence $b = (1, 12 - 5 - 6, 0) = (1, 1, 0)$. The sum of all six triangles has boundary equal to the sum of the six boundary edges, that is, inner loop + outer loop. So the two loops differ by a boundary and are homologous.

B31. (a) A closed connected surface with $\chi = 6 - 12 + 8 = 2$: $b = (1, 0, 1)$. (b) $n = (6, 12, 7)$, $\chi = 1$. It is a disk: $b_0 = 1$, and $b_2 = 0$. Indeed, a 2-cycle cannot contain a triangle with an edge lying in only one triangle, which excludes the three neighbors of the removed triangle; and if a triangle is excluded, so are its neighbors across edges lying in two triangles. By connectedness every triangle is excluded. So $b_1 = b_0 + b_2 - \chi = 0$, and $b = (1, 0, 0)$. (c) $n = (7, 12, 6)$, $\chi = 1$; it is a cone, so $b = (1, 0, 0)$.

B32. (a) $n = (7, 12, 8)$, $\chi = 3$; $b_0 = 1$, $b_2 = 2$ (each sphere gives a 2-cycle, and they are independent since they share no triangle), so $b_1 = b_0 + b_2 - \chi = 0$. (b) $n = (6, 11, 8)$, $\chi = 3$; again $b_0 = 1$, $b_2 = 2$, $b_1 = 0$.

B33. Torus: $\chi = 0$, $b = (1, 2, 1)$. Torus minus a triangle: $n = (9, 27, 17)$, $\chi = -1$, and $b_2 = 0$ (a 2-cycle would have to contain all triangles by propagation, but one is missing), so $b_1 = 1 + 0 + 1 = 2$ and $b = (1, 2, 0)$. Genus 3: $\chi = -4$, $b = (1, 6, 1)$. Five projective planes: $\chi = -3$, $b = (1, 5, 1)$.

B34. $\epsilon = 0.6$: the hexagon, $b = (1, 1)$. $\epsilon = 0.9$: the octahedron, $b = (1, 0, 1)$ (by Proposition 6.14 and $\chi = 2$). $\epsilon = 1$: the full simplex, $b = (1, 0, 0, 0, 0, 0)$.

B35. $\chi = 7 - 15 + 9 = 1$, so $b_1 = b_0 + b_2 - \chi = 1 + 0 - 1 = 0$. The answer $b_1 = 2$ would give $b_0 - b_1 + b_2 = -1 \neq \chi$.

B36. Every edge $\{a, b\}$ with $1 \notin \{a, b\}$ satisfies $\{a, b\} = \partial_2\{1, a, b\} + \{1, a\} + \{1, b\}$. So every 1-cycle is homologous to a cycle made of edges through 1, and such a sum can only be a cycle if it is empty (each $\{1, a\}$ contributes the vertex a, which no other edge through 1 contains). Hence $b_1 = 0$. With $n = (5, 10, 10)$, $\chi = 5$, and $b_2 = \chi - b_0 + b_1 = 4$.

Part C

C1. If $y \in B_r(x)$, let $s = r - d(x, y) > 0$. For $z \in B_s(y)$, $d(x, z) \leq d(x, y) + d(y, z) < r$, so $B_s(y) \subseteq B_r(x)$. If U, V are open and $x \in U \cap V$, choose $r_1, r_2 > 0$ with $B_{r_1}(x) \subseteq U$, $B_{r_2}(x) \subseteq V$; then $B_{\min(r_1,r_2)}(x) \subseteq U \cap V$.

C2. Let $f : X \to Y$ be continuous with X compact, and let $\{V_\alpha\}$ be an open cover of $f(X)$ by open sets of Y. The sets $f^{-1}(V_\alpha)$ are open and cover X, so finitely many of them, $f^{-1}(V_{\alpha_1}), \ldots, f^{-1}(V_{\alpha_n})$, cover X. Then $V_{\alpha_1}, \ldots, V_{\alpha_n}$ cover $f(X)$.

C3. Suppose $f : S^1 \to \mathbb{R}$ is a homeomorphism, and let $p \in S^1$. Then f restricts to a homeomorphism $S^1 \setminus \{p\} \to \mathbb{R} \setminus \{f(p)\}$. But $S^1 \setminus \{p\}$ is an open arc, which is connected, while $\mathbb{R} \setminus \{f(p)\}$ is the union of two disjoint non-empty open half-lines. Alternatively: S^1 is compact and $\mathbb{R}$ is not.

C4. Count pairs (edge, triangle containing it): each triangle has 3 edges, and each edge lies in exactly 2 triangles (Lemma of Week 2), so $3F = 2E$. Substituting $F = \frac{2}{3}E$ into $V - E + F = \chi$ gives $V - \frac{E}{3} = \chi$, so $E = 3(V - \chi)$ and $F = \frac{2}{3}E = 2(V - \chi)$. For the torus, $E = 3V \leq \binom{V}{2}$ gives $6 \leq V - 1$, so $V \geq 7$.

C5. Triangulate S_1 and S_2, remove the interior of one triangle from each, and glue the two boundary triangles. Removing the two interiors lowers F by 2. Gluing identifies 3 pairs of vertices and 3 pairs of edges, lowering V and E by 3 each, which does not change $V - E$. Hence $\chi(S_1 \# S_2) = \chi(S_1) + \chi(S_2) - 2$.

C6. Singletons $\{i\}$ with $U_i \neq \emptyset$ are simplices, and if $\tau \subseteq \sigma$ is non-empty, then $\bigcap_{i \in \tau} U_i \supseteq \bigcap_{i \in \sigma} U_i \neq \emptyset$, so τ is a simplex.

C7. If $\sigma \in \check{C}_\epsilon(X)$, there is a point z with $d(x, z) \leq \epsilon$ for all $x \in \sigma$. For $x, y \in \sigma$, $d(x, y) \leq d(x, z) + d(z, y) \leq 2\epsilon$, so $\sigma \in VR_\epsilon(X)$.

C8. See Proposition 3.6 of the notes. If $\mathrm{st}(\sigma)$ is a subcomplex, the faces of $\sigma \in \mathrm{st}(\sigma)$ must contain σ, so σ has no proper faces: $\sigma = \{v\}$. If v lay in some $\tau \neq \{v\}$, the non-empty face $\tau \setminus \{v\}$ of $\tau \in \mathrm{st}(v)$ would not contain v, a contradiction.

C9. It suffices to check a single simplex σ. Every face of σ obtained by deleting two vertices v_i, v_j occurs twice in $\partial\partial\sigma$ (delete v_i first, or v_j first), so its coefficient is $1 + 1 = 0$. Hence $\partial_{k-1}\partial_k = 0$, and if $c = \partial_{k+1}d$ then $\partial_k c = 0$.

C10. See Theorem 6.12. Vertices joined by a path are homologous (their sum is the boundary of the path), so $b_0 \le c$. The parity map $C_0 \to (\mathbb{Z}/2)^c$ vanishes on boundaries and sends one chosen vertex per component to the standard basis, so $b_0 \ge c$.

C11. Let $r_k = \operatorname{rank} \partial_k$, with $r_0 = r_{d+1} = 0$. By rank–nullity, $\dim Z_k = n_k - r_k$ and $\dim B_k = r_{k+1}$, so $b_k = n_k - r_k - r_{k+1}$. In $\sum_k (-1)^k (r_k + r_{k+1})$ each r_k appears with opposite signs and cancels, so $\sum (-1)^k b_k = \sum (-1)^k n_k$.

C12. See Proposition 6.14. There are no 3-simplices, so $b_2 = \dim Z_2$. A 2-cycle is a set S of triangles meeting every edge an even number of times. Since every edge lies in exactly two triangles, with each triangle S contains its neighbors across all its edges, and by connectedness all triangles. So $Z_2 = \{0, \text{all triangles}\}$ and $b_2 = 1$. With $b_0 = 1$, Euler–Poincaré gives $b_1 = 2 - \chi$.

C13. The chain groups and boundary matrices of $K \sqcup L$ are direct sums (block diagonal matrices), so ranks, cycles and boundaries all add.

Part D

D1. Graphs cannot distinguish a hollow triangle from a filled one, and cannot represent group interactions such as a paper with three authors; simplicial complexes can, and their homology detects holes in all dimensions. Abstract complexes can be stored and checked exactly as lists of integer sets, without floating point intersection tests, and they exist for data without coordinates (distance matrices) or for spaces that do not embed in low dimensions.

D2. Up to dimension 2 the program builds the boundary of the tetrahedron and reports $\chi = 4 - 6 + 4 = 2$, a fake sphere. The correct complex is the full tetrahedron with $\chi = 1$. To get correct information about holes of dimension k, the complex must be built up to dimension $k + 1$.

D3. Octahedron, edge $\{N, 1\}$: $\operatorname{lk}(N) \cap \operatorname{lk}(1) = \{2, 4\} = \operatorname{lk}(\{N, 1\})$; the condition holds and the edge can be contracted, giving a 5-vertex sphere. Tetrahedron, edge $\{1, 2\}$: $\operatorname{lk}(1) \cap \operatorname{lk}(2)$ contains the edge $\{3, 4\}$, which is not in $\operatorname{lk}(\{1, 2\}) = \{3, 4\}$; contracting would merge two triangles and destroy the surface.

D4. See the proposition in Section 4: the circumcenter lies in the triangle, the segments to the midpoints divide it into three quadrilaterals, each with all corners within the circumradius R of its vertex, and $R \le r/\sqrt{3}$ by Jung's theorem. For sensors, it means that a region filled by triangles of the Rips complex built from the communication graph is covered, so holes in coverage can only appear where the Rips complex has holes.

D5. $3 \cdot 69{,}451 = 208{,}353 = 2 \cdot 104{,}288 - 223$. $\chi = 34{,}834 - 104{,}288 + 69{,}451 = -3 = 2 - 2g - b$ with $b = 5$, so $g = 0$. Capping adds 5 vertices and 223 edges and triangles, so χ becomes 2: a sphere.

D6. Consecutive atoms are $12 \sin 15° \approx 3.11$ Å apart, so the wall is closed already for $R = 1.7$. The atoms at 30° and 330° are 6 Å apart, so their inflated disks meet at $R = 3$, closing the mouth. The center is at distance 6 from all atoms and is the last

point to be covered, at $R = 6$. So there is a cavity for $3 \leq R < 6$, that is, for probe radii $1.3 \leq \rho < 4.3\,\text{Å}$.

D7. The four neighbors of a failed interior sensor form a 4-cycle in which no triangle exists, since the only vertex adjacent to all of them has failed. A failed corner sensor leaves no cycle around the missing region, because the region is on the outside of the network. The corner point is at distance 1 from the nearest working sensor, and $1 > \sqrt{2}/\sqrt{3} \approx 0.82$, so it is not covered, although the Rips complex does not detect this.